

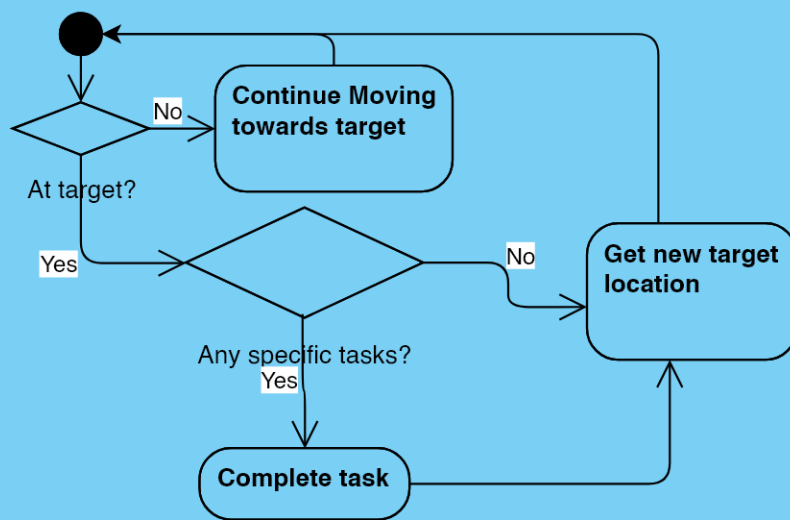
Technical Design Document

Algorithms:

1. Pathfinding for each enemy, such as a direct beeline towards the player, or finding a spot on the map to reach instead.
2. Algorithm required for the director to spawn enemies, increasing the price of certain enemies to ensure they aren't too common compared to others
3. Random Algorithm for picking upgrades for the player, being able to select from a list of applicable upgrades that the player can branch off from and be updated with new upgrades.

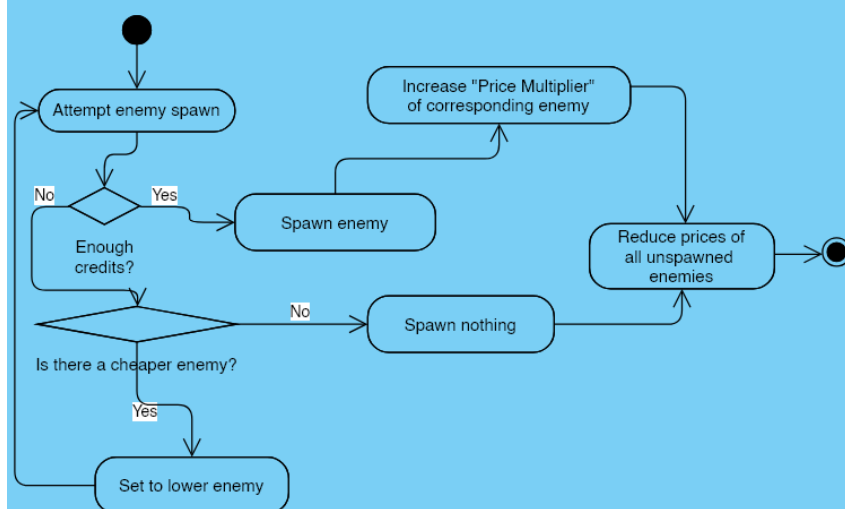
Pathfinding:

Pathfinding

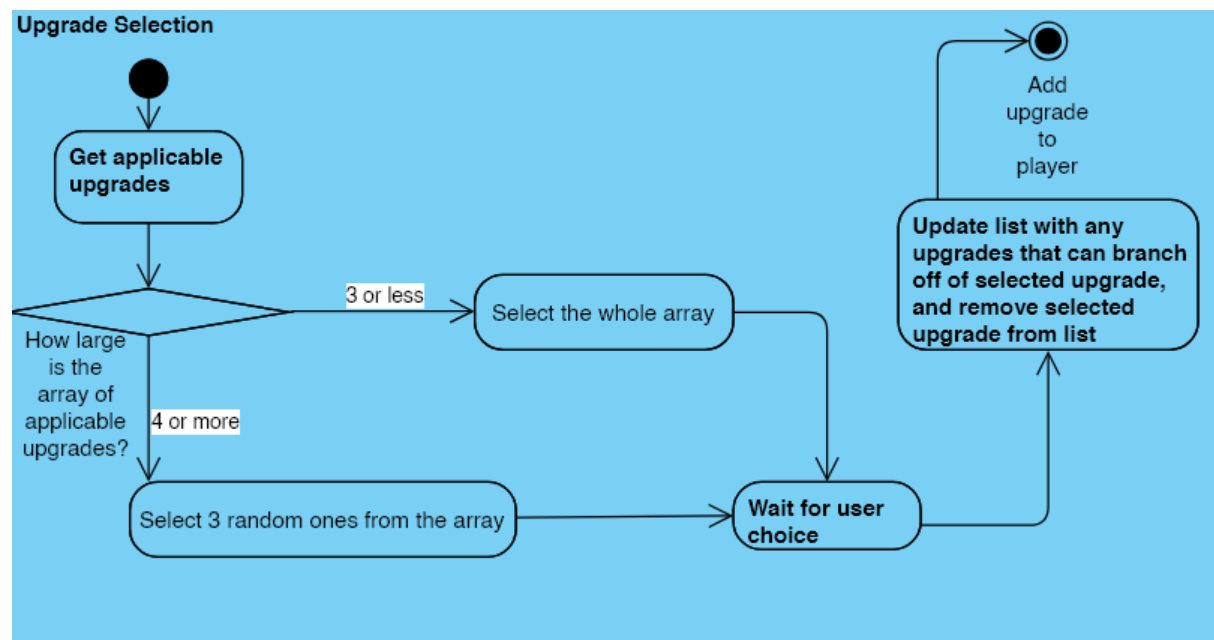


Director Algorithm:

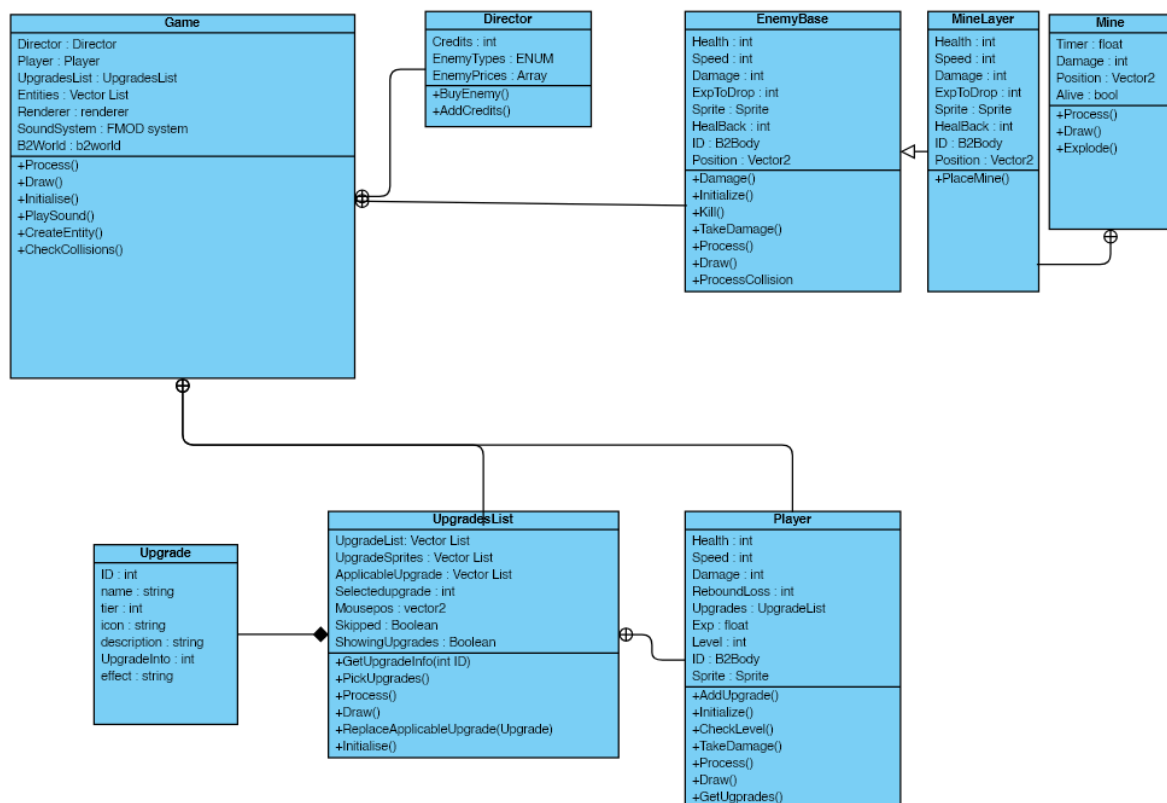
Spawn enemy



Upgrade Selection:



UML Class Diagram:



Debug Features:

1. Player info
 - Health
 - Damage
 - Momentum
 - Positions of various items (mouse, player)
 - Velocity
2. Cheats
3. Enemy info
 - Type
 - Damage
 - Health
4. Scene selector (splash screen and game)
5. Director info
 - Enemy costs
 - Credits

Coding Standards:

1. Pointers start with m_p for ease of distincting
2. Names are consistent with their function
3. Comments are used to keep functions and variables understandable
4. All enemy related files are stored in enemies folder in assets, upgrades in upgrade folder.

Relevant file formats:

1. OTF, TTF for fonts
2. Png for sprites
3. Wav for sound
4. .cpp and .h files for code
5. Lib,dll for libraries and other external resources

Acceptance Test:

1. Can the game open?
2. Can the game restart?
3. Can the player kill enemies?
4. Does levelling up work?
5. Do upgrades actually apply to the player?
6. Does the director spawn enemies?
7. Can the player leave the screen space?
8. Will the player be able to move quickly?
9. Does the game render everything in proper order?
10. Are all objects affected by deltatime and move consistently?